

Skills Summary

Modeling Discrete Linear Change and Arithmetic Sums

Use this reference while completing your worksheet or when studying.

Skill 1 — Turning a situation into a term rule

Set up first, compute second. Say in words what n counts (chairs, weeks, rows) and what a_n measures, with its unit. A situation that changes by the same amount at every stage is arithmetic.

Term rule: $a_n = a_1 + (n - 1)d$. The $n - 1$ is there because stage n is $n - 1$ steps past stage 1.

Careful: a one-off charge that is *not* repeated (a joining fee) sits outside the step: that total is $a_1 + dn$, not a term of the sequence.

Example: A ladder costs \$25 for the first day of hire and \$14 for each day after. What does a 6-day hire cost?

Step 1. The daily charge is constant after the first day, so the cost by day n is arithmetic with $a_1 = 25$ and $d = 14$.

Step 2. Six days is five extra days: $25 + 5(14) = 25 + 70$.
cost = **95** dollars

Try it: A tool costs \$40 for the first day and \$12 for each day after. What does a 9-day hire cost?

check: 981 dollars

Skill 2 — Finding the stage number

Rule: when the question gives you a *value* and asks *when*, set the term rule equal to that value and solve for n .

$$a_1 + (n - 1)d = \text{target} \implies n - 1 = \frac{\text{target} - a_1}{d} \implies n = \frac{\text{target} - a_1}{d} + 1.$$

Sanity check: a stage number must be a whole number. A fraction means the target is never hit exactly — report the first stage past it, and say so.

Example: For $a_n = 7 + 5(n - 1)$, which term equals 62?

Step 1. The value is given and the stage number is unknown, so this is an equation to solve, not a substitution.

Step 2. $7 + 5(n - 1) = 62$, so $5(n - 1) = 55$ and $n - 1 = 11$.
 $n = \mathbf{12}$

Try it: For $a_n = 100 - 8(n - 1)$, which term equals 44?

check: 8 = u

(!) Watch out: do not forget the +1 at the end. Dividing gives the number of *steps*; the stage number is one more than that.

Skill 3 — Adding up a finite run

Rule: $S_n = \frac{n}{2}(a_1 + a_n)$. Pair the smallest term with the largest, the next-smallest with the next-largest, and every pair has the same total.

Order of work: find a_n first, then average it with a_1 , then multiply by how many terms there are — and n is a count of terms, not a count of steps.

Example: For $a_n = 6 + 4(n - 1)$, find the sum of the first 9 terms.

Step 1. A total over a fixed run of stages is a finite arithmetic sum, so pair the ends rather than adding nine numbers.

Step 2. Last term: $6 + 8(4) = 38$.

Step 3. Nine terms, so $\frac{9}{2}(6 + 38) = 9(22)$.

$$S_9 = 198$$

Try it: For $a_n = 3 + 7(n - 1)$, find the sum of the first 10 terms.

check: $S_{10} = 345$